

Active Beauty Agefinity™ The 3D Y reshaper



Crafted by white and green technologies



Focus on the product

Ageing impact on mitochondrial metabolism and reticular dermis organisation

Mitochondria are unique organelles which are the power generators of cells, converting oxygen and nutrients into adenosine triphosphate (ATP), the chemical energy that allows cell metabolism. They are able to synthesise proteins independently from the cell nucleus and cytoplasm. They represent an important factor in human evolution by supporting essential cell functions (energy powerhouse, protection of DNA, signaling of cell reproduction, activation of cell apoptosis, maintenance of cell electrochemical integrity...)

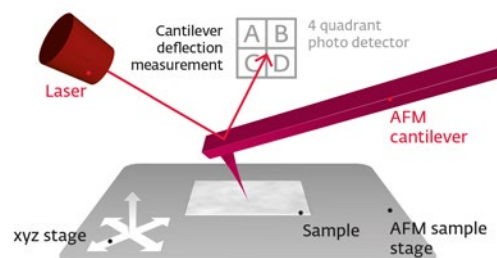
The **reticular dermis** is the lower layer of the dermis, found under the papillary dermis, composed of dense irregular connective tissues and featuring densely packed collagen fibres. **These protein fibres give the dermis its properties of strength, extensibility, and elasticity.** The orientation of collagen fibres within the reticular dermis creates lines of tension called Langer's lines, which are essential for wound healing and mechanical properties.

A decline in mitochondrial activity has been associated with normal cell ageing¹. With age and external stresses, some damages are done to the mitochondria of dermal fibroblasts, leading to Reactive Oxygen Species (ROS) production and inadequate energy production in the process to convert oxygen into energy. As a consequence, the genes expression pattern of dermal fibroblasts will be changed, causing a disturbance in collagen production, inflammation², disruption of detoxification pathway (proteasome) and oxidation of skin pigments (lipofuscin), leading to **disorganisation of the extracellular matrix (ECM), premature skin ageing and apparition of age spots.**

An innovative technique to characterise the complex dermis structure

A change of production pattern of collagen in the ECM, caused by any of the phenomena described before, will lead to a disorganisation of the reticular dermis, changes in packing of the fibres and will lead to a decline of mechanical properties in the skin.

The complex hierarchical structure of collagen can be evaluated by using an innovative methodology: Atomic Force Microscopy (AFM). This technique scans the surface of skin cells or skin sections, using a sharp tip which will identify and follow the relief of the support during the scanning. This method therefore gives information about biomechanical properties and topography of the studied skin.



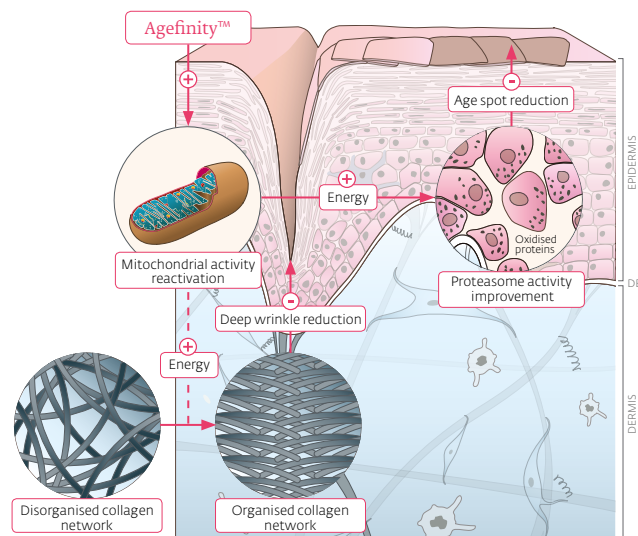
Agefinity™: a green energy source for the cells, to reshape ageing skins

Mannose-6-Phosphate is a human metabolite used to generate energy in our cells and to drive proper functions. It is an essential component to promote glycolysis in the cell (fostering energy production).

In order to replenish the energy of the cells, Active Beauty has crafted a Mannose-6-Phosphate complex, from a vegetal source, using green chemistry: Agefinity™ which will:

- ▶ Reprogram **mitochondrial energy metabolism** in older cells.
- ▶ Promote **skin matrix restructuring**.
- ▶ Reinforce **Dermo Epithelial Junction (DEJ)**.
- ▶ **Reorganise skin matrix** similar to younger skin.
- ▶ Significantly **reduce visible age spots**.
- ▶ Visibly **reduce crows feet wrinkles**.
- ▶ Reduce **neck wrinkles to remodel the Y-shape** of the face.

Agefinity™ is the perfect solution to act on the lower layer of the dermis to reshape the Y-zone (face and neck).



1- Nuo Sun, Richard J. Youle, and Toren Finkel. The Mitochondrial Basis of Aging. Mol Cell. 2016 Mar 3; 61(5): 654–666.

2- Jean Krutmann, Peter Schroeder. Role of Mitochondria in Photoaging of Human Skin: The Defective Powerhouse Model. Volume 14, Issue 1, August 2009, Pages 44-49

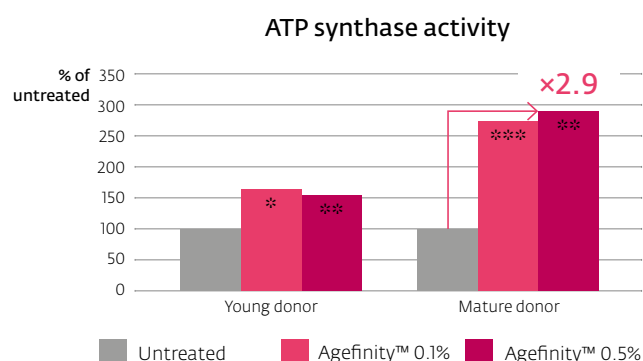
Biological activity

Reactivating mitochondrial energy metabolism (*in vitro*)

1. Action on mitochondrial target

Keratinocytes (NHEK) from a young donor's skin (26 year old) and from an older donor's skin (51 year old), were treated with 0.1% and 0.5% of Agefinity™ for 72 hours or left untreated.

Results: Agefinity™ at 0.1% and 0.5% significantly **increase the cell metabolism** by increasing the activity of the ATP synthase, with 2.9 times increase of metabolism at 0.5% on older donor cells compared to untreated.

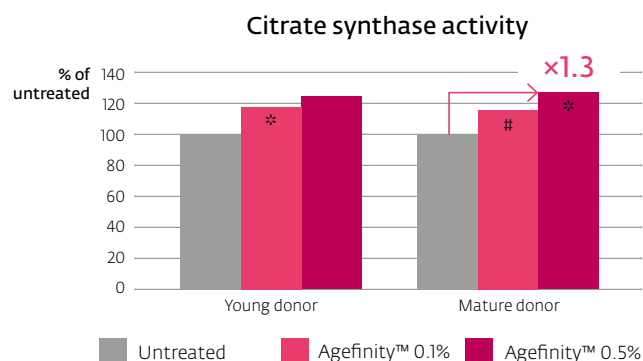


2. Action on mitochondrial biogenesis

Skin cells from a young donor's skin (26 year old) and from an older donor's skin (51 year old), were treated with 0.1% and 0.5% of Agefinity™ for 72 hours or left untreated.

The citrate synthase enzymatic activity was evaluated using spectrofluorometer to evaluate the mitochondrial biogenesis.

Results: Agefinity™ at 0.1% and 0.5% **significantly increase the mitochondrial biogenesis** in older cells 1.3 times more than untreated.



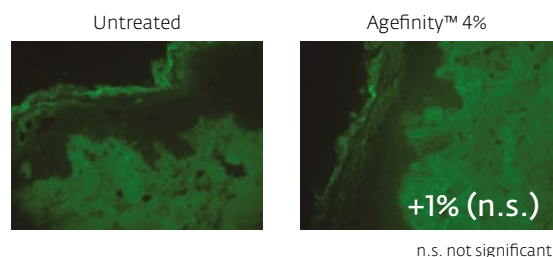
Student's t-test vs untreated: #p<0.1, *p<0.05, **p<0.01, ***p<0.001

A sugar with no impact on protein glycation (*ex vivo*)

Skin explants were treated with 4% of Agefinity™ or left untreated for 48 hours.

The glycation (bonding of a sugar molecule to a protein or lipid molecule without enzymatic regulation that damages proteins) was evaluated using specific immunohistochemistry with anti-AGE primary antibody.

Results: Agefinity™ **doesn't induce any protein glycation** due to the exposition to sugar.



n.s. not significant

Biological activity

Skin matrix restructuration

1. Transcriptomic analysis (*in vitro*)

Fibroblasts from a 28 year old donor were treated with 4% of Agefinity™ for 6 hours. The modulations of genes expression were evaluated using RT-qPCR on total extracted RNA.

CTGF: Connective tissue growth factor (regulator of collagen synthesis)

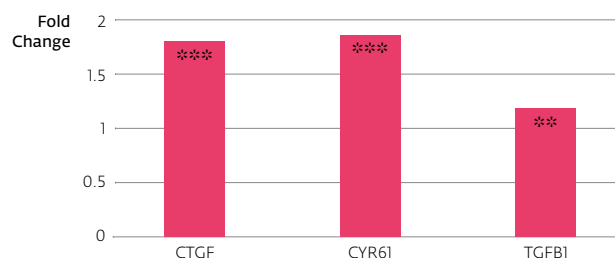
CYR61: Cysteine-rich angiogenic inducer 61 (extracellular matrix component essential for fibroblast migration, dermis regeneration and restructuration)

TGF-β1: Transforming growth factor beta 1 (stimulates proliferation of fibroblasts)

Results: Agefinity™ stimulates the expression of several genes involved in the control of dermis restructuration.

Student's t-test: **p < 0.01, ***p < 0.001

Impact on dermis restructuration genes



2. Proteomic analysis (*ex vivo*)

Skin explants from a 59 year old Caucasian donor were treated with 4% of Agefinity™ for 7 days or left untreated. The variation of protein expressions were quantified using immunostaining quantification after specific immunohistochemistry.

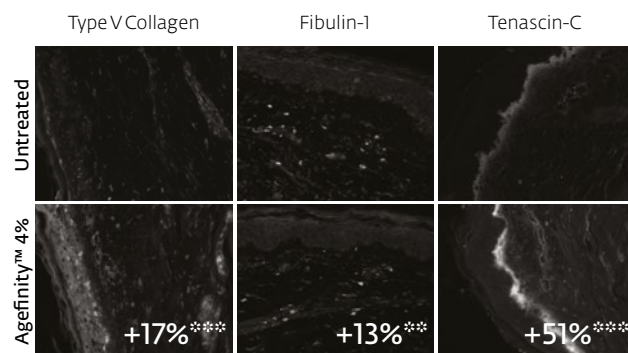
Type V Collagen: Fibrillary collagen, minor component of the dermal matrix which associates with collagen type I

Fibulin-1: Component of the extracellular matrix essential to maintain structure of the DEJ

Tenascin-C: Matrix protein involved in the cell cohesion at the DEJ located at the basal layer.

Results: Agefinity™ stimulates the expression of various proteins involved in dermis structuration.

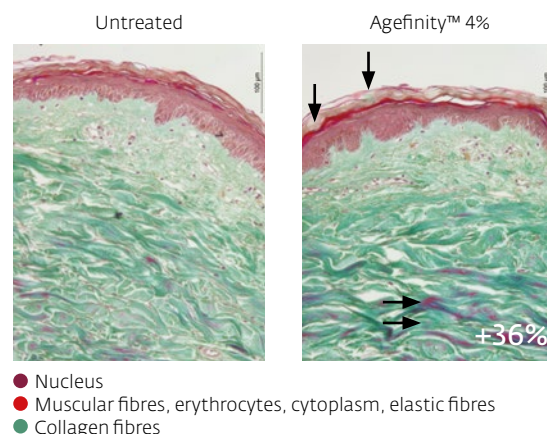
Student's t-test: **p < 0.01, ***p < 0.001



3. Morphological analysis (*ex vivo*)

Skin explants from a 59 year old Caucasian donor were treated with 4% of Agefinity™ for 7 days or left untreated. Collagen fibres and muscular tissues were stained using Masson's Trichrome staining.

Results: Agefinity™ at 4% visibly increases the density of reticular dermis by +36%, the DEJ is better defined and the epidermis is more cohesive.



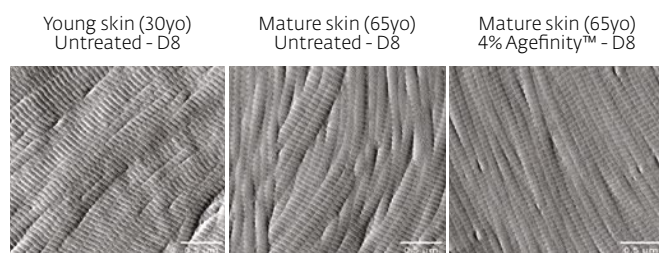
Biological activity

4. Collagen fibres organisation (*ex vivo*)

Skin explants from 2 donors (30 and 65 year old Caucasian women) were topically treated every day with 4% of Agefinity™ for 8 days or left untreated. The organisation of collagen fibres into young and mature explants was evaluated using Atomic Force Microscopy (AFM) and Small-angle X-ray scattering (SAXS).

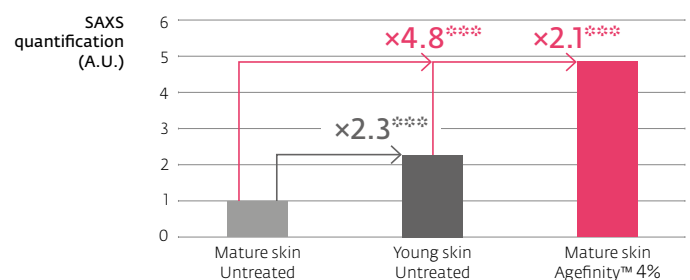
Results: Agefinity™ at 4% improves collagen fibres cohesion and organises the dermis up to 4.8 times better than untreated mature skin, and up to 2.1 times better than untreated young skin.

AFM measurements on skin explants



Student's t-test: ***p < 0.001

Dermis organisation improvement



Control of protein ageing (*ex vivo*)

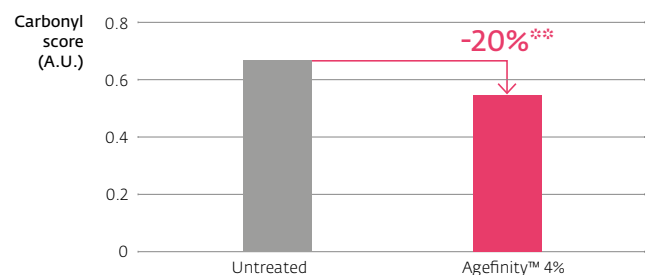
1. Increase of proteasome activity

Skin explants from a 59 year old Caucasian woman were treated with 4% of Agefinity™ for 7 days or left untreated. The proteasome activity was evaluated by labelling carbonylated proteins (oxidised proteins) with a fluorescent probe.

Results: Agefinity™ at 4% significantly reduces the carbonyl score by -20% compare to untreated reflecting a **significant reactivation of proteasome activity in a mature skin**.

Student's t-test: **p < 0.01

Restoring proteasome activity (decrease of oxidised proteins content)

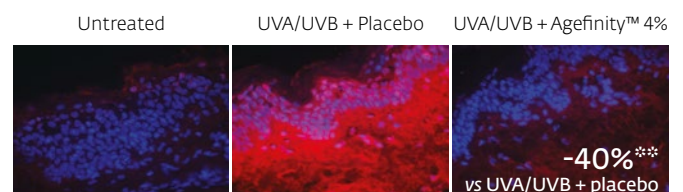


2. Reduction of oxidised proteins content

Skin explants from a 45 year old Caucasian woman were pre-treated with 4% of Agefinity™ or Placebo or left untreated for 24 hours. The skin explants were then exposed to UVA radiation (8J/cm²) and UVB radiations (200mJ/cm²) and treated again with the products for 24 hours. The protein oxidation level was then measured using immunostaining.

Results: Agefinity™ at 4% induces a significant reduction of oxidised proteins production under UV radiations, down to -40% compared to the placebo.

Student's t-test: **p < 0.01



Efficacy

Anti-ageing efficacy of Agefinity™ (clinical test)

To evaluate the anti-ageing benefits of Agefinity™ in face care applications, two clinical tests were conducted in double blind versus placebo.

Study 1 (#1, #2 and #3): carried out on 22 women (average age 59 year old) presenting crows feet wrinkles and nasolabial wrinkles. Volunteers applied the formula containing 4% Agefinity™ or the placebo (same formula without active) on hemi-face twice a day for 56 days.

Study 2 (#4): performed on 2 groups of 20 volunteers (average age 56 year old) presenting neck wrinkles. The efficacy of formula containing 4% Agefinity™ relative to placebo (same formula without active) was evaluated on full face after an application twice a day for 56 days.

1. Reduction of visible age spots

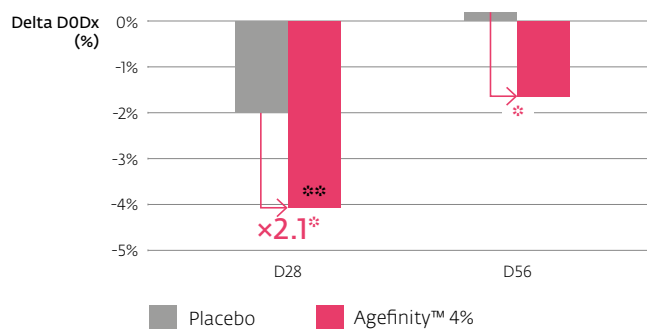
The number of visible spots was evaluated using the VISIA CR 2.3® to realise digital photography of the face at D0, D28 and D56.

Results: After 28 days, Agefinity™ **significantly reduces the number of visible age spots** compared to placebo, with a reduction up to 2.1 times more efficient. The effect is still visible and significant versus placebo after 56 days.

Student's t-test vs placebo: *p< 0.05

Student's t-test vs D0: **p< 0.01

Decrease of visible age spots number

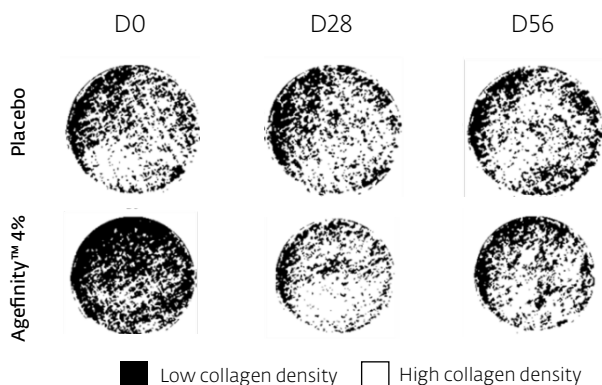
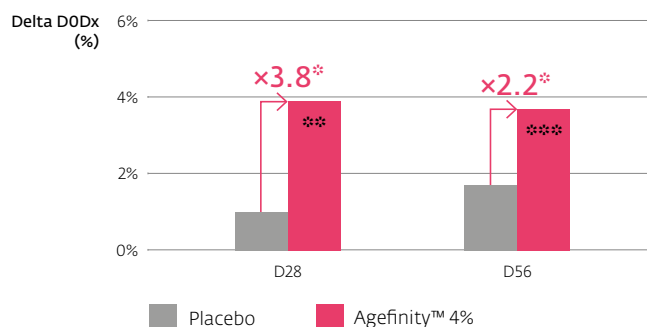


2. Improvement of skin collagen organisation

The density and distribution of collagen were analysed at the commissure level using SIAscope® device and SIAMetrics® analysing system after 28 and 56 days.

Results: Agefinity™ **significantly stimulates relative collagen density** compared to placebo with an increase up to 3.8 folds. The **collagen is better organised** after only 1 month.

Increase in collagen density



Student's t-test vs Placebo: * p<0.05

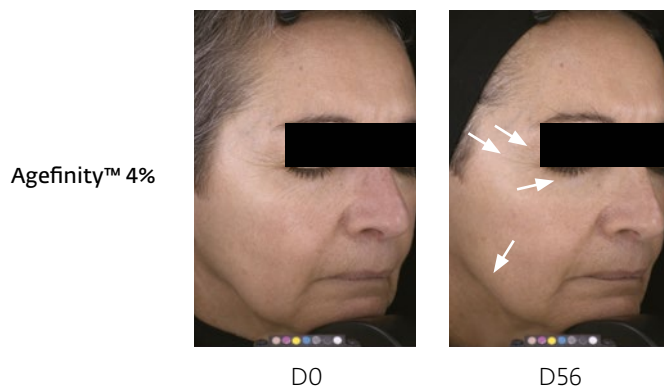
Student's t-test vs D0: ** p<0.01, *** p<0.001

Efficacy

3. Anti-wrinkle efficacy

The number of wrinkles was evaluated on the crow's feet area using the VISIA CR 2.3® to realise digital photography of the face at D0, D28 and D56.

Results: Agefinity™ at 4% induces a significant reduction of crow's feet wrinkle number, down to -11.5% versus placebo. A visual improvement of the bottom part of the face (jowls) can also be noticed on the captures pictures.



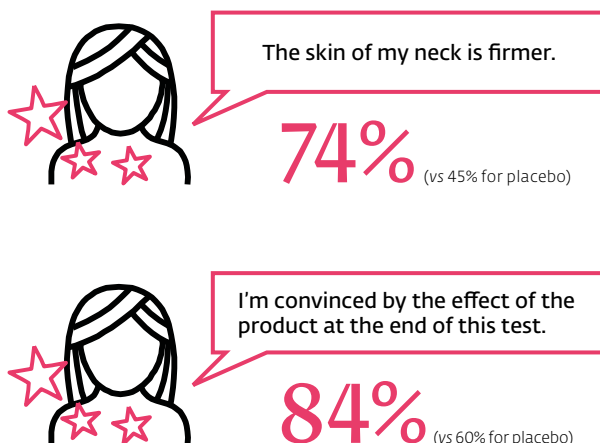
Wilcoxon test vs placebo: * p<0.05

4. Reduction of neck wrinkles

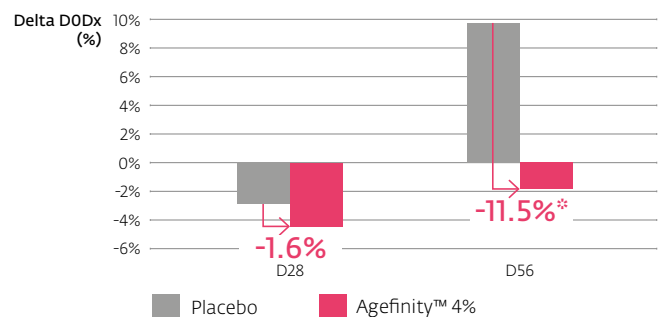
Neck wrinkles properties (volume and depth) were evaluated on the neck area using AEVA HE® at D0, D28 and D56. Panelists were also asked to assess both products efficacy on their neck after 56 days of use.

Results: Agefinity™ induces a significant reduction of the neck wrinkles volume and depth, respectively by up to -32.5% and -8.3% versus placebo, as confirmed by the captured images.

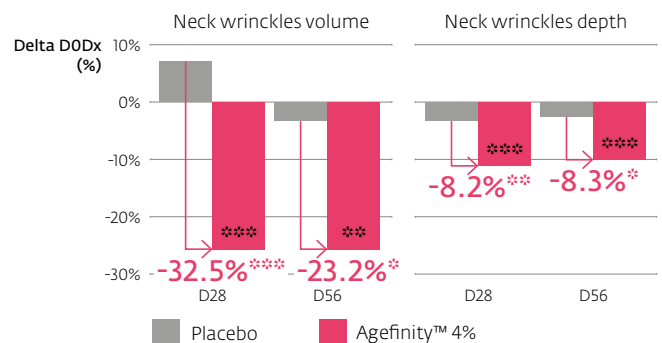
Wilcoxon test vs D0, Mann Whitney test vs placebo: *p<0.05, **p<0.01, ***p<0.001



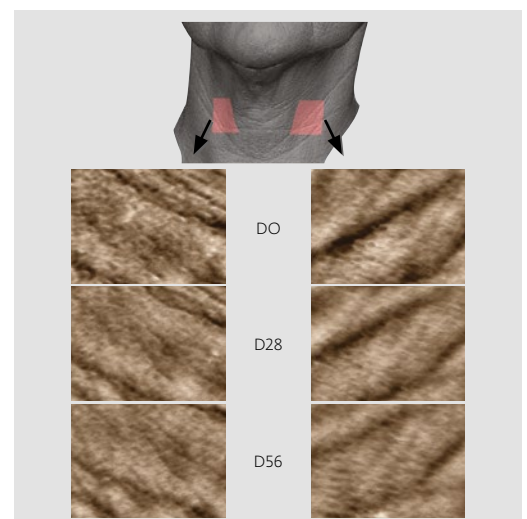
Reduction of crow's feet wrinkles



Reduction of neck wrinkles



Illustrative pictures of Agefinity™ 4% on the neck wrinkles



Summary



Technical information

Suggested INCI:	Glycerin (and) Water (and) Sodium Mannose Phosphate (and) Mannose
Origin:	Plant & biotechnology
Preservation:	Preservative free
Appearance:	Clear liquid
Solubility:	Soluble in water
Dosage:	0.1% to 4%
Processing:	Can be added to the water phase at the end of the formulation process, at a pH between 2 and 8.

Claims

Claims:	Anti-wrinkle, collagen organiser, skin lifting, anti-age spots, anti tech-neck, anti text-neck, remodeling the V-shape of the face, improvement of the Y-zone.
Applications:	Anti-ageing serums or creams, anti-ageing products for the reduction of deep wrinkles, intensive treatments for facial rejuvenation, nasolabial anti-wrinkles products, anti teck-neck products, anti-crow's feet products, anti-age spots serums, skincare for mature skin, dermocosmetics.

Givaudan Active Beauty Sales Offices

Europe

Givaudan France SAS
19-23 rue de la Voie des Bans
FR-95100 Argenteuil (France)

Givaudan UK Ltd
Magna House
76-80 Church Street
Staines, TW18 4XR (United Kingdom)

Naturex SA
250 rue Pierre Bayle - BP 81218
84911 Avignon Cedex 9 (France)

global.cosmetic@givaudan.com

Asia Pacific

Givaudan Singapore Pte Ltd
1 Pioneer Turn
627576 Singapore (Singapore)

Givaudan Shanghai Ltd
298 Li Shi Zhen Road
Pudong Zhang Jiang High Tech Park
201203 Shanghai (China)

Latin America

Givaudan do Brasil Ltda
Av. Eng^o Billings - 2185, Edifício 31, 1^a Andar - Jaguaré
05321-010 São Paulo - SP (Brazil)

North America

Givaudan Fragrances Corp.
40 W - 57th Street - Floor 17
NY 10019 - New York (United States)

The data in this document ("Data"): (i) has been prepared by Givaudan in accordance with Givaudan's internal protocols and procedures; (ii) is provided to Customer for its information and internal use only; (iii) is provided without warranty of any kind, including, without limitation, any implied warranty of accuracy, merchantability, fitness for particular purpose or non-infringement of third party intellectual property rights. In no event shall Givaudan be liable to Customer or any third party for any losses, indemnities or damages of any kind (including, without limitation, any and all direct, special, indirect, incidental, or consequential damages or lost profits or revenues) that may arise out of, or in connection with, the use of the Data by Customer. Customer is solely responsible for assessing the accuracy and reliability of the Data for its own purposes (including, without limitation, Customer's end-use applications), and assumes all risks and liabilities arising out of or in connection with the use of the Data. LEAFLET-AGEFINITY-04.19

www.givaudan.com